

Teacher Guide — middle / module-09-subtraction-nikhilam

IKS Curriculum

Table of Contents

Overview

Module 9 — Subtraction by Complements (*Nikhilam Sūtra*)

Age band: Middle · **Grades:** 6–8 · **Duration:** 10 sessions × 45 min ($\approx$ 2 weeks)

Learning objectives

By the end of this module, students will be able to:

1. **Recite and translate** the Nikhilam sūtra — *Nikhilam Navataścaramam Daśataḥ*, “all from nine and the last from ten.”
2. **Find the complement (*pūraka*)** of any number to the next power of ten — 10, 100, 1000, 10 000 — without paper computation.
3. **Subtract from a power of ten** (e.g. $1000 - 567$) by reading off the digit-wise complement, with no borrowing.
4. **Apply** the same idea to general subtractions (e.g. $8243 - 567$) by combining a complement with a place-value adjustment.
5. **Explain algebraically** why the rule works — that $10^n - A = (10^n - 1) - A + 1$, and that “ $(10^n - 1) - A$ ” is exactly the digit-wise subtraction from nines.
6. **Compare** the speed and error-pattern of Nikhilam subtraction with standard borrowing.

NEP 2020 alignment

- **§4.6** — Integration of Indian Knowledge Systems (Vedic mathematics) with mainstream arithmetic.
- **§4.23** — Experiential learning via paired drills, mental-math games, and shopkeeper role-play.
- **§4.27** — Local context: connects to real-world change-counting and bill-rounding.
- **§5.1** — Foundational numeracy and mental-arithmetic fluency.

Prerequisites

- Confident standard-algorithm subtraction with borrowing across at least two columns.
- Place-value fluency for 3- and 4-digit numbers.
- “Make 10” reflex from Module 5 helps — the Nikhilam rule reuses it heavily.

Key vocabulary

See `_shared/glossary.md` for full entries.

- *Nikhilam Navataścaramam Daśataḥ* — “all from nine and the last from ten”
- *pūraka* — complement (the number that completes another to a base)
- *sūtra* — short formula or aphorism
- *vyavakalana* — subtraction
- *Tirthaji* — Bharati Krishna Tirthaji (1884–1960), compiler of the 16 sūtras

Materials checklist (whole module)

- Whiteboards / scrap paper for live drills
- Stopwatch or wall clock (one per pair is enough)
- A class wall-poster of the Nikhilam sūtra in IAST + English
- Pretend currency (paper notes denominated in 10, 100, 500, 1000) for the shopkeeper activity
- Printed practice sheets — complement drills, subtraction drills, mixed sets
- Optional: a calculator (used *only* for self-checking)

Day-by-day overview

Day	Title	Focus	Assessment
1	The Sūtra and the Hook	Meet <i>Nikhilam</i> ; baseline subtraction timing	Baseline time
2	Complement to 100	Visual + verbal complements of 2-digit numbers	Recall check
3	Complement to 1000	Pattern recognition; speed drill	Formative quiz
4	Subtracting from a Power of Ten	$1000 - N$, $10\,000 - N$ with no borrowing	Activity rubric
5	Mid-module Check	Speed test + formative	Formative quiz
6	Why It Works	Algebraic justification + bar-	Reflection paragraph

Day	Title	Focus	Assessment
		model proof	
7	Two's Complement Connection	Binary subtraction in computers uses the same idea	Cross-disciplinary note
8	General Subtraction with Complements	When and how to combine	Mixed problem set
9	Speed Practice + Method Comparison	Same problems, both methods	Comparison table
10	Final Assessment + Project Share	Summative quiz + 3-min presentations	Summative + project

Citation sources used

- **SutraGanita corpus** (/Users/sree/Projects/SutraGanita/content/) — primary references in `sources.md`:
 - Tirthaji, B. K. *Vedic Mathematics* (1965) — the foundational text listing the 16 sūtras; *Nikhilam* is sūtra #2.
 - *Microsoft Word - sutras1.pdf* — single-page sūtra list, used for the wall poster and Day 1 reading.
 - *FUNDAMENTAL AND VEDIC MATHEMATICS.pdf* — chapter on complement-based subtraction with worked examples.
- See `sources.md` for the full citation chain and honest provenance notes.

Project brief (one-line teaser)

Each pair makes a **shopkeeper's quick-change chart**, a **60-second teaching video** for Grade 4, OR a **two's-complement explainer poster** that connects Nikhilam to binary arithmetic. Full brief in `assessments/project-brief.md`.

10-Day Lesson Plan

Module 9 (Middle) — 10-Day Lesson Plan

At a glance

Day	Session title	Lesson file	Activity / assessment
1	The Sūtra and the Hook	<code>days/day-01-hook.md</code>	Baseline timed test
2	Complement to 100	<code>days/day-02-complement-100</code>	Pair drills

Day	Session title	Lesson file	Activity / assessment
		.md	
3	Complement to 1000	days/day-03-complement-1000.md	Formative quiz Part A
4	Subtracting from a Power of Ten	days/day-04-power-of-ten.md	activities/activity-01-complement-drill.md
5	Mid-module Check	days/day-05-mid-module-check.md	Formative quiz Part B
6	Why It Works	days/day-06-why-it-works.md	Reflection paragraph
7	Two's Complement Connection	days/day-07-twos-complement.md	Cross-disciplinary note
8	General Subtraction with Complements	days/day-08-general-subtraction.md	activities/activity-02-shopkeeper.md
9	Speed Practice + Comparison	days/day-09-speed-comparison.md	Comparison table
10	Final Assessment + Project Share	days/day-10-final-assessment.md	quizzes/summative.md + project

Pacing notes

- **Day 1** establishes a baseline: students time themselves on $10000 - 4567$ using standard borrowing. **Day 5** retests the same problem with the Nikhilam method. The visible speedup is the module's motivation.
- **Days 2–4** are the core skill: finding complements and applying them to power-of-ten subtraction. If the class is solid by Day 3, Day 4's drill can be extended.
- **Day 6** ("Why it works") and **Day 7** (two's complement) can be reordered — some classes find the binary parallel more motivating than the algebra; in that case do Day 7 first, then loop back to algebra on Day 8 morning.
- **Days 8–9** introduce the general case and revisit speed. Day 10 is summative.

Daily warm-up (recurring)

Every class starts with **1 minute of "complement to 10" fluency**: teacher calls out a digit; students call back the partner that makes 10. ($8 \rightarrow 2$, $6 \rightarrow 4$, $9 \rightarrow 1$, etc.) Same warm-up as Module 5 — it is the prerequisite reflex for Nikhilam.

What students leave the module with

- Instant recall of complements to 10, 100, and 1000 for any number under 1000.
- Confidence subtracting from a power of ten without writing a borrow.
- An algebraic statement (in their own words) of why the method works: $10^n - A = (10^n - 1) - A + 1$, and the digit-wise rule applies because $10^n - 1$ is “all nines.”
- One real-world application — change-counting, deficit checks, or a binary parallel.
- One project artefact they can share with a younger student.

Teacher Notes

Teacher Notes — Module 9 Nikhilam Subtraction (Middle)

Posture

The win in this module is **fluency**: a student looks at $10\ 000 - 4567$ and writes 5433 without a single borrowed digit. The Sanskrit name *Nikhilam Navataścaramaṇ Daśataḥ* is the cultural anchor; the *technique* is what they take home — and it is genuinely faster than standard borrowing once practised.

Honesty about provenance: the Nikhilam *sūtra* is real and printed in Tirthaji’s 1965 book. The claim that it derives from a Vedic appendix is unverified and disputed by mainstream Sanskritists. Don’t oversell this as “ancient.” At Middle band, name the *sūtra* and the date (1965), and move on. Senior students unpack the historicity in depth.

Common misconceptions

1. **“You can’t subtract a bigger digit from a smaller one — that’s why we borrow.”** The Nikhilam method *avoids* this entirely by subtracting each digit *from* 9 (and the last from 10). 9 is always bigger than any single digit, so no borrow can ever be needed. Demonstrate this on Day 2.
2. **“The complement is the answer to the subtraction.”** Only when subtracting from a clean power of ten. For $1000 - 567 = 433$, yes — 433 is the complement. For $1234 - 567$, no — you have to combine the complement with the leading digits. Day 8 fixes this.
3. **“The last digit is ‘special’ just because Tirthaji said so.”** No — 9, 9, 9, 9 would give you 9999, not $10\ 000$. The “+1” needed to make $9999 \rightarrow 10\ 000$ is exactly what “the last from ten” provides. Show this on Day 6.
4. **“This is binary two’s complement.”** Closely related, not identical. Two’s complement is “all bits flipped, plus 1.” Nikhilam is “all digits from 9, last digit from 10.” Same structure, different base. Be precise on Day 7.
5. **“If the subtrahend has fewer digits than the base, just pad with zeros.”** Yes — $1000 - 7$ means treat 7 as 007, complement digit-wise: $9-0, 9-0, 10-7 = 9, 9, 3 \rightarrow 993$. Common student error: forget to pad. Drill this on Day 4.

Differentiation hints

- **Tier up:** Generalise — find the complement of any number to any chosen base (not only powers of ten). E.g. complement of 38 to 50 is 12. Then subtract using the same idea.
- **Tier down:** Stick to complements to 100 until Day 4. Don't move to 1000 until the 100-complement is automatic.

Differential pacing

- A class with strong place-value baseline can compress to 7 days by combining Days 2+3 and Days 8+9.
- A class struggling with borrowing in the first place should NOT skip Day 2. The “complement to 100” drill builds the basic pattern; everything else stacks on it.

Sensitivity notes

- Some students are anxious about timed math. The timed activities are **personal baselines** — students race themselves, not each other. No public leaderboards.
- Avoid the framing “the West has slow borrowing, India has the fast method.” Standard borrowing works fine and is itself partly of Indian origin (Brahmagupta). Both methods are useful; Nikhilam is *especially* useful for subtractions from a round base.
- The Sanskrit sūtra is printed in IAST throughout. Do not transliterate ad-hoc (“nikhilam navataschcaramam”). Use the IAST form.

Project oversight

- Three project formats (shopkeeper's quick-change chart, 60-second teaching video, two's-complement explainer poster) each have different production challenges.
- Shopkeeper chart: confirm by Day 8 that the pair has chosen a real shop scenario (e.g. “you sold a 273-rupee item with a 1000-rupee note — how much change?”) and not an arbitrary list of numbers.
- Video projects: check by Day 8 that the pair has access to a phone with a camera AND a quiet place to record.
- Two's-complement poster: this is the most challenging artefact. Allow only pairs who have engaged seriously with Day 7. Strong differentiation choice.

Citation discipline

Citations for this module are book references (e.g. *Tirthaji, Vedic Mathematics (1965)*, *sūtra Nikhilam*) and PDF references (e.g. *FUNDAMENTAL AND VEDIC MATHEMATICS .pdf*, Nikhilam chapter). Students should cite at least one source in their project. This is the standard for math modules in this curriculum.

If you only have 6 days

- Day 1 (hook + baseline + complement to 10)
- Day 2 (combined: complements to 100 and 1000)
- Day 3 (subtracting from power of ten + mid-module retest)
- Day 4 (why it works — pick *either* algebra *or* two's complement, not both)

- Day 5 (one project session)
- Day 6 (final assessment, light project presentation)

What to flag for the curriculum lead

- If most students still struggle with the *base-padding* idea by Day 5, the Module 5 “make-10” reflex was not fully established. Worth a 10-minute reteach.
- If students adopt the method but can’t articulate why it works on Day 6, push the algebra harder before Day 7. Two’s complement only lands cleanly when the algebra has clicked.

Sources

Sources — Module 9 Nikhilam Subtraction (Middle)

The **vidya-karana RAG** does not cover Vedic-math sources. This module draws directly from SutraGanita’s source corpus (/Users/sree/Projects/SutraGanita/content/) and from the published Tirthaji corpus.

Primary source

Tirthaji, Bharati Krishna. *Vedic Mathematics* (1965), Motilal Banarsidass.

- Sūtra #2 of the 16: ***Nikhilam Navataścaramam Daśataḥ*** — “all from nine and the last from ten.”
- The sūtra is used in Tirthaji’s book for both subtraction (the focus of this module) and a multiplication trick (out of scope here; covered in a future module).
- Cite as: *Tirthaji, Vedic Mathematics (1965), sūtra Nikhilam (#2)*.

SutraGanita corpus sources

- **Microsoft Word - sutras1.pdf** — single-page listing of all 16 Tirthaji sūtras with one-line glosses. Used for the wall poster and the Day 1 reading. Available in /Users/sree/Projects/SutraGanita/content/.
- **FUNDAMENTAL AND VEDIC MATHEMATICS .pdf** — comprehensive teacher reference; the chapter on Nikhilam includes worked subtraction examples that we adapt for Days 4–5. Cite as: *FUNDAMENTAL AND VEDIC MATHEMATICS .pdf*, Nikhilam chapter.
- **Vedic Mathematics.pdf** — Tirthaji’s original text (scanned). Source of the verse *Nikhilam Navataścaramam Daśataḥ* as Tirthaji prints it.
- **Vedic Mathematics - Sixteen Mathematical Formula from the Vedas.pdf** — alternative published version with worked examples that match Tirthaji’s notation.

Historical and critical context (teacher reference)

- **Plofker, Kim.** *Mathematics in India* (2009), Princeton University Press. Background on classical Indian arithmetic and the broader history into which Tirthaji’s 20th-century synthesis fits. No verse from Plofker is quoted to students at Middle band.

- **Dani, S. G.** “Myths and Reality: On Vedic Mathematics,” *Frontline* (1993), reprinted in *Resonance* (October 2001). The scholarly critique of Tirthaji’s “Vedic” provenance claim. Senior-band reading; Middle-band teachers should know it exists.

Honest provenance note (for the teacher)

The Nikhilam *sūtra itself* is a real entry in Tirthaji’s 1965 list. Tirthaji presents it as Vedic; no other Sanskritist has produced a manuscript of the *Atharva-veda Pariśiṣṭa* he claims to be drawing from. Treat the *sūtra* and its method as a **20th-century synthesis** that may well draw on older Indian mental-arithmetic traditions but whose direct Vedic attribution is unverified.

What is *definitely* true:

- The complement-based subtraction technique (in some form) has parallels in earlier Indian arithmetic and in 19th–20th-century European systems (e.g. Trachtenberg).
- The Sanskrit phrasing and the systematic packaging into 16 *sūtras* is Tirthaji’s contribution.
- The method works. The history is just younger than the name suggests.

At Middle band we teach the method enthusiastically, name the *sūtra* accurately, and don’t oversell its antiquity. The Senior band module unpacks the historicity question in detail.

What to tell students about the citation chain

When students cite a fact about the Nikhilam method in their project, the citation looks like:

Tirthaji, Vedic Mathematics (1965), sūtra Nikhilam.

When they cite a worked example from a workbook PDF, the citation looks like:

FUNDAMENTAL AND VEDIC MATHEMATICS .pdf, Nikhilam chapter.

When they cite the binary two’s-complement parallel (Day 7), they may cite any introductory computer-science text; we suggest:

Patterson & Hennessy, Computer Organization and Design (any edition), two’s-complement section.

This citation chain is the standard for math modules in this curriculum.

Day 1 — The Sūtra and the Hook

Module: Nikhilam Subtraction · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will record a personal baseline time for $10\ 000 - 4567$ using standard borrowing, will recite the Nikhilam *sūtra* in Sanskrit, and will have watched the method solve the same problem visibly faster.

Materials

- Printed baseline sheet: ONE subtraction problem ($10\ 000 - 4567 = ?$). One per student.
- Stopwatch or wall clock (one per pair is plenty).
- Whiteboard / large paper for the teacher demo.
- A printed Nikhilam sūtra poster — IAST + English — on the wall.

Vocabulary introduced today

- *Nikhilam Navataścaramam Daśataḥ* (IAST: *nikhilam navataścaramam daśataḥ*; lit. “all from nine and the last from ten”) — the sūtra we’ll learn this fortnight.
- *sūtra* (IAST: *sūtra*; lit. “thread, formula, aphorism”) — a short mnemonic rule. Tirthaji’s *Vedic Mathematics* gives 16 of them.

Lesson flow

Warm-up (5 min)

- Teacher: “How long does it take you to solve $10\ 000 - 4567$? Make a guess.”
- Take 3–4 guesses. Write the range on the board (e.g. “15 sec to 60 sec”).
- Quick make-10 reflex round: teacher calls digits; class chorus the partner. $8 \rightarrow 2$, $6 \rightarrow 4$, $9 \rightarrow 1$, $3 \rightarrow 7$, $5 \rightarrow 5$, $4 \rightarrow 6$, $7 \rightarrow 3$, $2 \rightarrow 8$.

Baseline timing (8 min)

- Hand out the baseline sheet face-down.
- Students write their name and predicted time on the back.
- On “Go,” flip and solve using whatever method they normally use (standard borrowing).
- Record their actual time (the wall clock is running).
- Mark answer; honest self-correction — they’re racing themselves over the module, not each other.

Meet the sūtra (5 min)

- Point to the wall poster. Teacher reads slowly:
Nikhilam Navataścaramam Daśataḥ “All from nine and the last from ten.”
- Class echoes 3 times: soft / normal / loud.
- Teacher: “That’s the whole rule. Six Sanskrit words. By Day 10 you’ll use it without thinking.”

Teacher demonstration (12 min) — the same baseline problem, the Nikhilam way

Write on the board:

$$\begin{array}{r} 1\ 0\ 0\ 0\ 0 \\ - \quad \quad 4\ 5\ 6\ 7 \\ \hline \end{array}$$

Narrate:

1. “We want to subtract 4567 from 10 000. That’s 10^4 — a clean power of ten. Perfect Nikhilam territory.”
2. “Treat 4567 as a 4-digit number. Apply the rule digit-by-digit.”

Step	Apply rule	Get
First digit (4):	“from 9”	$9 - 4 = 5$
Second digit (5):	“from 9”	$9 - 5 = 4$
Third digit (6):	“from 9”	$9 - 6 = 3$
Last digit (7):	“from 10”	$10 - 7 = 3$

3. Read the answer top-to-bottom: **5433**.
4. **Verify** (with the class): $4567 + 5433 = 10\ 000$. ✓
5. “Notice — I never borrowed. Not once. The ‘all from nine’ rule means we’re always subtracting from 9, which is bigger than any single digit.”

Activity (10 min) — First try

- Students re-do the same problem ($10\ 000 - 4567$) on the back of their baseline sheet, using the Nikhilam rule. Refer to the wall poster.
- Time themselves again.
- Compare: how much slower (or faster) was the Nikhilam method on the FIRST try?

Wrap-up (5 min)

- Show of hands: who was faster with Nikhilam on first try? Who was slower? (Most will be slower — that’s expected. Practice fixes it.)
- Preview Day 2: “Tomorrow we drill the foundational skill — finding the complement of a 2-digit number to 100, fast.”
- Set the wall-poster Sanskrit chant as the daily opener for the next 9 days.

Student-facing content

The Nikhilam Sūtra

Nikhilam Navataścaramam Daśataḥ — “all from nine and the last from ten.”

To subtract from 10, 100, 1000, 10 000, ...: 1. Subtract each digit *from 9*, **except the last digit** — subtract it *from 10*. 2. Read the answer top-to-bottom. 3. Verify: original + answer = the round base.

No borrowing. Ever.

Today’s baseline: - My time with my normal method = _____ seconds - My time with Nikhilam (first try) = _____ seconds

I expect the second number to drop by Day 5.

Homework

- Memorise the sūtra. Be able to say it from memory tomorrow.
- Practice: find the complement of these 2-digit numbers to 100:
 - 73, 41, 28, 86, 95, 17, 60, 52, 34, 89.
 - Time yourself. Aim for under 30 seconds for all ten.

Differentiation

- **Tier up:** Try Nikhilam on 100 000 – 23 456. Time it.
- **Tier down:** If the 4-digit problem feels overwhelming, drill complements to 100 (2-digit numbers) for the activity instead.

Teacher notes

- The baseline time is the *single most important* artefact of Day 1. Make sure every student records it honestly. Day 5's retest depends on it.
- Don't promise a specific speedup — say “you'll likely be faster by Day 5; how much depends on you.”
- The Sanskrit pronunciation is hard. Model it slowly; let students laugh on the first attempt. By Day 5 they'll all say it cleanly.
- One paraphrase from Tirthaji that helps: “Nikhilam (निखिलं) means ‘whole,’ ‘entire,’ ‘all.’ Navataḥ means ‘from nine.’ Caramaḥ means ‘the last.’ Daśataḥ means ‘from ten.’ Six words; one method.”

Citation(s) used in this lesson

- Tirthaji, *Vedic Mathematics* (1965), sūtra Nikhilam (#2). The full sūtra reads *Nikhilam Navataścaramaḥ Daśataḥ*.
- *Microsoft Word - sutras1.pdf* — wall-poster source.

Day 2 — Complement to 100

Module: Nikhilam Subtraction · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will find the complement (*pūraka*) of any 2-digit number to 100 in under 3 seconds, using the Nikhilam rule applied to two digits.

Materials

- Whiteboards or scrap paper for live drills
- A printed wall poster: 9, 9, 9, 9, 9, . . . 10 showing the digit-wise rule visually
- Pair-drill sheets (20 two-digit numbers each)
- Stopwatch (one per pair)

Vocabulary introduced today

- *pūraka* (IAST: pūraka; lit. “that which completes”) — the **complement**: the number you add to fill up to 100, 1000, etc.

Lesson flow

Warm-up (5 min)

- Sūtra chant 3 times.
- Make-10 reflex: 10 quick digit-partners.

Core (15 min) — The pattern

1. **Pose:** “What’s the complement of 73 to 100?” Take guesses.
2. Walk through it the Nikhilam way:

$$\begin{array}{r} 100 \\ - 73 \\ \hline \end{array}$$

- First digit (7): $9 - 7 = 2$
 - Last digit (3): $10 - 3 = 7$
 - Answer: **27**.
 - Verify: $73 + 27 = 100$. ✓
3. **Why “all from 9, last from 10”?** Show on the board:
$$99 + 1 = 100$$
 - If we subtract each digit *from* 9 we get the complement to **99**, not 100.
 - The “+1” needed to bridge $99 \rightarrow 100$ is exactly what “the last from 10” provides: $10 - x$ is the same as $(9 - x) + 1$.
 - That’s the whole trick. The last digit absorbs the “+1.” Every other digit just goes “from 9.”
 4. **Demo 4 more, quickly, with the class chanting along:**

Number	First digit	Last digit	Complement	Check
41	$9 - 4 = 5$	$10 - 1 = 9$	59	$41 + 59 = 100$ ✓
28	$9 - 2 = 7$	$10 - 8 = 2$	72	$28 + 72 = 100$ ✓
86	$9 - 8 = 1$	$10 - 6 = 4$	14	$86 + 14 = 100$ ✓
95	$9 - 9 = 0$	$10 - 5 = 5$	05 = 5	$95 + 5 = 100$ ✓

5. **Edge case: trailing zero.** What is the complement of 30? Apply the rule: $9 - 3 = 6$ (for tens), $10 - 0 = 10$ (for units). But 10 isn’t a digit — it carries. So you get $6 \mid 10 \rightarrow 70$.

Easier: notice that the last *non-zero* digit gets the “from 10” treatment; trailing zeros stay as zeros.

Or more simply: $30 + 70 = 100$, so complement of 30 is 70. The rule still works, you just respect place value.

Activity (20 min) — Paired drill

- Pairs face each other across desks. One holds the drill sheet (20 two-digit numbers); the other says complements out loud.
- Round 1: 60 seconds. Count correct answers.
- Swap roles. Round 2.
- Round 3: swap to a new sheet (different numbers). Aim to beat your Round 1 count.

Wrap-up (5 min)

- Quick poll: who got 15+ in 60 sec? 18+? 20?
- Recall paraphrase: “*To complete to 100, take each digit from 9; the last digit takes from 10.*”
- Preview Day 3: “*Tomorrow — same idea, but complement to 1000.*”

Student-facing content

Complement (*pūraka*) to 100

For any 2-digit number AB:

- Tens digit: $9 - A$
- Units digit: $10 - B$

Examples: - 73 $\rightarrow$ 27 ($9-7=2$, $10-3=7$) - 41 $\rightarrow$ 59 ($9-4=5$, $10-1=9$) - 95 $\rightarrow$ 05 ($9-9=0$, $10-5=5$)

Verify by adding: number + complement = 100.

Tirthaji’s *Vedic Mathematics* (1965) gives the rule as *Nikhilam Navataścaramam Daśataḥ* — and a “complement to 100” is just the rule applied to two digits.

Homework

- Memorise complements to 100 of the multiples of 5: 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95. These should become automatic.
- Time yourself: 20 random complements in under 60 seconds. Practise twice.

Differentiation

- **Tier up:** Try complements to 200, 300, 500. The rule modifies — find complement to 100, then adjust. (Tier-up students should be able to articulate that adjustment by the end of class.)
- **Tier down:** Use the rule only when both digits of the number are non-zero. Trailing zeros will be drilled on Day 3.

Teacher notes

- Most common error today: $9 - 0 = 9$ (correct) but $10 - 0 = 10$ (cannot be a digit). Students will write 0 and lose 10. Reinforce: if the last digit is 0, the complement's last digit is also 0, and the "10" gets absorbed into the next column. Or: just note that the rule cleanly handles non-zero last digits; trailing zeros are a quick edge-case.
- Many students try to do this in their head WITHOUT going digit-by-digit. That's fine if they get the right answer — but make sure they can *show* the digit-wise method when asked. Day 6's "why it works" depends on the digit-wise framing.
- Pair students with different reflexes. The faster-reflex student often models; the slower benefits.

Citation(s) used in this lesson

- Tirthaji, *Vedic Mathematics* (1965), sūtra Nikhīlam (#2) — the source of the digit-wise rule.

Day 3 — Complement to 1000

Module: Nikhīlam Subtraction · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will find the complement of any 3-digit number to 1000 in under 4 seconds, and will recognise the same pattern extends to 10 000, 100 000, and beyond.

Materials

- Whiteboard
- Pair-drill sheets (20 three-digit numbers; second set with mixed 3- and 4-digit)
- Stopwatch
- Calculator (for self-verification only)

Vocabulary

- *pūraka* — complement (introduced Day 2; reuse)

Lesson flow

Warm-up (5 min)

- Sūtra chant 3 times.
- Quick fire: teacher calls 2-digit numbers; students chorus the complement to 100.
 - $73 \rightarrow 27$, $41 \rightarrow 59$, $28 \rightarrow 72$, $86 \rightarrow 14$, $95 \rightarrow 05$, $60 \rightarrow 40$, $17 \rightarrow 83$.

Core (15 min) — Three digits, same rule

1. **Pose:** "What's the complement of 738 to 1000?"
2. Walk through:

$$\begin{array}{r} 1000 \\ - 738 \\ \hline \end{array}$$

- First digit (7): $9 - 7 = 2$
 - Second digit (3): $9 - 3 = 6$
 - Last digit (8): $10 - 8 = 2$
 - Answer: **262**.
 - Verify: $738 + 262 = 1000$. ✓
3. “Notice — same six-word rule. We just have one more ‘from 9’ step in the middle.”
4. **The general pattern:**

For an n -digit number, the complement to 10^n is:

$$(9 - d_1), (9 - d_2), \dots, (9 - d_{n-1}), (10 - d_n)$$

where $d_1 \dots d_n$ are the digits left-to-right.

5. **Demo 4 more, with class chanting along:**

Number	Working	Complement	Check
419	9-4, 9-1, 10-9 = 5, 8, 1	581	$419 + 581 = 1000$ ✓
256	9-2, 9-5, 10-6 = 7, 4, 4	744	$256 + 744 = 1000$ ✓
803	9-8, 9-0, 10-3 = 1, 9, 7	197	$803 + 197 = 1000$ ✓
999	9-9, 9-9, 10-9 = 0, 0, 1	001 = 1	$999 + 1 = 1000$ ✓

6. **Quick extension to 10 000:** Complement of 2347 to 10 000?

- 9-2, 9-3, 9-4, 10-7 = **7653**.
- $2347 + 7653 = 10\ 000$. ✓
- “Same rule. Just more digits.”

Activity (15 min) — Speed drill

- Pair-drill: 20 three-digit numbers. Aim for all 20 in under 90 seconds.
- Self-check using a calculator: number + your complement should equal 1000.
- Second sheet: 10 four-digit numbers, complement to 10 000. Aim under 60 seconds for the set.

Wrap-up + Formative quiz (10 min)

- Formative quiz Part A from quizzes/formative.md — 5 questions on slips of paper, 5 min.
- Collect. Next class includes the result.

Student-facing content

Complement to 1000 (and beyond)

For any 3-digit number ABC: - First digit: **9 – A** - Second digit: **9 – B** - Last digit: **10 – C**

Same six-word rule. Same approach for 4-digit numbers (one extra “from 9”) and 5-digit numbers (two extra “from 9”s).

Examples: - 738 → 262 - 419 → 581 - 2347 → 7653 (complement to 10 000)

“Nikhilam navataścaramam daśataḥ — *all from nine and the last from ten.*” (Tirthaji, *Vedic Mathematics*, 1965, sūtra #2.) The “all” handles every digit except the last; “the last from ten” handles the last digit’s special role.

Homework

- Find the complement of each of these to 1000:
 - 156, 423, 789, 802, 64 (treat as 064), 7 (treat as 007), 555, 900.
- Find the complement of each of these to 10 000:
 - 3456, 1027, 8888, 75 (treat as 0075), 999.
- Time yourself. Total time goal: under 2 minutes.

Differentiation

- **Tier up:** Complement of 5-digit numbers to 100 000. Time goal: under 6 sec each.
- **Tier down:** Stay on 3-digit complements. The four-digit and five-digit cases come naturally once 3-digit is automatic.

Teacher notes

- The most common error: a number like 64 has only two digits; students forget the rule wants a 3-digit number, so they apply the rule to “64” (giving 36) instead of “064” (giving 936). Make sure they “pad with zeros” mentally.
- Another common error: the last digit being 0. For 800 → complement to 1000: digit-wise gives 9-8, 9-0, 10-0 = 1, 9, 10. The “10” carries. Easier reading: 800 + 200 = 1000, so complement is 200. The rule still works (1, 9, 10 → 1, 10, 0 → 200), but it’s a place where the padding/carry interacts. Use this as a teachable moment; preview Day 6’s algebraic justification.
- Some students will sniff out a shortcut: “the complement just makes everything add to 9 except the last column which adds to 10.” Great — that *is* the rule, restated. Praise.

Citation(s) used in this lesson

- Tirthaji, *Vedic Mathematics* (1965), sūtra Nikhilam (#2).
- *FUNDAMENTAL AND VEDIC MATHEMATICS .pdf*, Nikhilam chapter — worked 3-digit and 4-digit examples.

Day 4 — Subtracting from a Power of Ten

Module: Nikhilam Subtraction · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will compute subtractions of the form $10^n - A$ (e.g. $1000 - 567$, $10\,000 - 4783$) directly by reading off the complement of A , with no written borrow.

Materials

- Whiteboard
- `activities/activity-01-complement-drill.md` worksheet (printed)
- Stopwatches (one per pair)

Vocabulary

(All terms reused from Days 1–3.)

Lesson flow

Warm-up (5 min)

- Sūtra chant.
- Quick fire: complements to 100, then complements to 1000. Teacher calls; students chorus.

Core (10 min) — The shift from “complement” to “subtraction”

1. **The pivot.** Write on the board:

$$1000 - 567 = ?$$

“We’ve spent two days finding the complement. The complement of 567 to 1000 is 433. Notice that’s also the answer.”

2. **Why?** *“If $A + B = 1000$, then $1000 - A = B$ and $1000 - B = A$. Complement IS the difference, when you’re subtracting from the base.”*
3. **Worked examples, with class predicting:**

Problem	Working	Answer
$1000 - 567$	9-5, 9-6, 10-7 = 4, 3, 3	433
$1000 - 248$	9-2, 9-4, 10-8 = 7, 5, 2	752
$10\,000 - 4567$	9-4, 9-5, 9-6, 10-7 = 5, 4, 3, 3	5433
$10\,000 - 89$	9-0, 9-0, 9-8, 10-9 = 9, 9, 1, 1	9911 (pad 89 → 0089)
$100\,000 - 23\,456$	9-2, 9-3, 9-4, 9-5, 10-6 = 7, 6, 5, 4, 4	76\,544

4. **The padding rule.** “If A has fewer digits than the base 10^n , pad A with leading zeros. 89 in $10000 - 89$ becomes 0089 . The Nikhilam rule needs as many digits as the base has zeros.”

Activity (20 min) — Complement Drill (timed pairs)

See [activities/activity-01-complement-drill.md](#).

- 8 problems on a sheet. Mixed bases: 4 are $1000 - X$, 3 are $10\ 000 - X$, 1 is $100\ 000 - X$.
- Pair self-correction. Time the pair.

Wrap-up (10 min)

- Compare with standard borrowing: pick one problem (say $10\ 000 - 4567$) and ask half the class to solve by borrowing, half by Nikhilam. Same time limit. See who finishes first.
- Discussion: “When does Nikhilam beat standard borrowing? When does standard borrowing match it?” (Answer: Nikhilam dominates when subtracting from a round base; matches roughly when subtracting from a non-round number — that’s the Day 8 topic.)
- Preview Day 5: “Tomorrow — mid-module retest of the Day 1 baseline. Same problem. Different speed.”

Student-facing content

Subtracting from a Power of Ten

To compute $10^n - A$:

1. Make sure A has exactly n digits (pad with leading zeros if needed).
2. Apply the Nikhilam rule digit-by-digit: each digit “from 9,” last digit “from 10.”
3. Read off the answer.

No borrowing required. Verify by adding: original + answer = base.

Tirthaji’s sūtra (1965): *Nikhilam Navataścaramaṁ Daśataḥ*. Six words; an entire subtraction shortcut.

Homework

- 12 problems on the back of today’s worksheet. Time yourself. Try to finish all 12 in under 4 minutes.
- One reflection sentence: “How did the Nikhilam method feel today compared to Day 1?”

Differentiation

- **Tier up:** Mixed-base problems where the base is given symbolically. E.g. let $B = 10^6 = 1\ 000\ 000$; compute $B - 234\ 567$.
- **Tier down:** Stick to $1000 - A$ for the homework. Step up to $10\ 000$ only when the 1000 case is automatic.

Teacher notes

- This is the day students go “ohh — that’s why we learned the complement.” Make the connection explicit. The complement IS the answer when subtracting from the base.
- A student will inevitably ask: “*What if the problem is $9999 - 567$? Does it still work?*” Answer: yes, but without the “last from 10” — the answer is $9999 - 567$, computed digit-wise as $9-5, 9-6, 9-7 = 432$. Try it: $567 + 432 = 999$. ✓. So $9999 - 567 = 9432$. This is the special case where ALL digits are “from 9” (because there’s no “+1” needed when the base is $10^n - 1$). Good for Tier-up.
- Avoid letting students treat the method as a black box. Insist they verify at least one of their answers by adding.

Citation(s) used in this lesson

- Tirthaji, *Vedic Mathematics* (1965), sūtra Nikhīlam (#2).
- *FUNDAMENTAL AND VEDIC MATHEMATICS .pdf*, Nikhīlam chapter — provided the 5-digit example pattern.

Day 5 — Mid-module Check

Module: Nikhīlam Subtraction · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will retest the Day 1 baseline problem using the Nikhīlam method, record the new time, and complete the formative quiz Part B that consolidates Days 1–4.

Materials

- The Day 1 baseline sheets (returned today)
- Fresh paper for the retest
- quizzes/formative.md Part B (printed)
- Stopwatch

Lesson flow

Warm-up (5 min)

- Sūtra chant 3 times (loud).
- Make-10 reflex: 10 quick partners.

Retest of Day 1 baseline (8 min)

- Hand back the Day 1 sheets so students can see their old time.
- Give a fresh sheet with the same problem: $10\ 000 - 4567 = ?$
- “*Solve it the Nikhīlam way. Time yourself.*”
- Record new time below the old time.
- **% improvement** = $(\text{Day 1 time} - \text{Day 5 time}) / \text{Day 1 time} \times 100$. Compute on the spot.

Brief sharing (5 min)

- 3 volunteers share their % improvement.
- *“Some of you went from 30 sec to 8 sec. Some from 18 sec to 15 sec. Both are wins. The point is the trend.”*

Formative quiz Part B (15 min)

- Hand out quizzes/formative.md Part B.
- 12 mixed problems: 3 complement-to-100, 4 complement-to-1000 / $1000 - N$, 3 $1000 - N$, 1 four-digit padding case, 1 short answer (“state the sūtra in English and Sanskrit”).
- Closed notebook. Aim to finish in 12 minutes.
- Collect for grading.

Diagnostic discussion (10 min)

- While the quiz is being graded (or based on a quick teacher scan), call out three common error types on the board:
 1. **Padding error** — student wrote $10000 - 89 = 9921$ because they treated 89 as “8 9” not “0 0 8 9.” Reteach: pad with leading zeros to match the base’s digit count.
 2. **Last-digit error** — student wrote $9 - 7 = 2$ for the units instead of $10 - 7 = 3$. Reteach: the LAST digit always takes from 10. Always.
 3. **Trailing-zero handling** — $1000 - 70$: digit-wise gives 9-0, 9-7, 10-0 = 9, 2, 10. The 10 carries: 9, 3, 0 → 930. Verify: $70 + 930 = 1000$. ✓. Acknowledge this is a corner case; offer the easier “see 70 as 070, complement directly” framing.
- Take questions.

Wrap-up (2 min)

- Project choice form due tomorrow. Three options: shopkeeper chart, teaching video, two’s-complement poster. (See [assessments/project-brief.md](#).)
- Preview Day 6: *“Why does this work? Tomorrow we prove it — with algebra.”*

Student-facing content

Mid-module Check

Day 1 time: ____ sec → Day 5 time: ____ sec → % change: ____

If you improved less than expected, the most common reason is one of three things:

1. Padding error — you forgot leading zeros.
2. Last-digit error — you used “from 9” for the last digit instead of “from 10.”
3. You’re still translating mentally between “borrow” and “complement.”

Don’t panic. Days 6–9 fix all three.

Homework

- Choose your project format (shopkeeper chart / video / poster) and write a 3-sentence plan.
- 5 mixed Nikhilam subtractions: $1000 - 358$, $10\ 000 - 6427$, $1000 - 72$, $100\ 000 - 12\ 345$, $10\ 000 - 9$. (Last one is 9991; check the padding!)

Differentiation

- **Tier up:** Try mixed-base, mixed-method speed test: 6 problems in 60 seconds. Use Nikhilam where helpful, standard where not.
- **Tier down:** Re-do the Day 4 worksheet. Don't take the quiz cold.

Teacher notes

- The retest is the emotional pivot of the module. Most students will see a real improvement; honor it without overselling. ("You're 50% faster — and you're still getting faster.")
- Use the formative quiz to triage: anyone scoring below 60% gets a 10-min pull-aside on Day 7 before the two's-complement lesson. Don't let them carry a foundational gap into the harder material.
- Distinguish: this is a *formative* check, not summative. No grade weight. Day 10 is summative.

Citation(s) used in this lesson

- Tirthaji, *Vedic Mathematics* (1965), sūtra Nikhilam (#2).

Day 6 — Why It Works

Module: Nikhilam Subtraction · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will explain *why* the Nikhilam rule produces the correct answer, using the identity $10^n - A = (10^n - 1) - A + 1$, and will draw a bar-model that shows the same idea visually.

Materials

- Whiteboard
- Worksheet: "explain to a friend" prompt + space
- Coloured pencils for the bar model
- The 16-sūtra list (Microsoft Word - sutras1.pdf) visible

Lesson flow

Warm-up (5 min)

- Sūtra chant.
- Quick fire: 4 complements (mix of 2-digit and 3-digit).
- Return graded formatives. Acknowledge the common errors.

Core (20 min) — The mathematical reason

1. **The core identity (write boxed on the board):**

$$10^n - A = (10^n - 1) - A + 1$$

2. **Why this identity?** “ 10^n is 1 followed by n zeros — e.g. $1000 = 10^3$. $10^n - 1$ is the same number minus one — e.g. 999. That’s just ‘all nines.’ So the identity says: subtracting A from a round base equals (subtracting A from all-nines) plus 1.”
3. **Why does that help?** “Subtracting any digit from 9 is trivial — there’s no borrow. We always have $9 \geq \text{digit}$. So $(10^n - 1) - A$ is a clean digit-by-digit subtraction with no borrow. Then we add 1 at the end.”
4. **Where does the “+1” go?** “Into the last digit. So instead of doing ‘from 9’ on the last digit and then ‘+1,’ we just do ‘from 10’ on the last digit. Same thing. That’s why the sūtra says ‘all from nine and **the last from ten.**’”
5. **Walk through a concrete example with full algebra:**

$$\begin{aligned} &1000 - 567 \\ &= (1000 - 1) - 567 + 1 && \leftarrow \text{identity} \\ &= 999 - 567 + 1 \\ &= (9-5)(9-6)(9-7) + 1 && \leftarrow \text{digit-wise from 9 – no borrow!} \\ &= \begin{array}{r} 4 \\ 3 \\ 2 \end{array} + 1 \\ &= 432 + 1 \\ &= 433. && \checkmark \end{aligned}$$

Now compare with the sūtra direct version:

$$\begin{aligned} &1000 - 567 \\ &= (9-5)(9-6)(10-7) && \leftarrow \text{Nikhilam, last from 10} \\ &= \begin{array}{r} 4 \\ 3 \\ 3 \end{array} \\ &= 433. && \checkmark \end{aligned}$$

“Same answer. The sūtra is just the identity, made into a mnemonic.”

6. **Bar model.**

Draw a long horizontal bar labelled 1000. Inside, mark off a left segment labelled 567 and a right segment labelled ? (the unknown). “What we want is the right segment.”

Now ask: “What’s the length of the bar minus 1?” — Answer: 999.

Mark a tiny +1 sliver at the right end of the bar. Now $999 = 567 + (\text{right segment} - 1)$, so right segment $= (999 - 567) + 1 = 432 + 1 = 433$.

“That’s the picture-version of the algebra. Same thing.”

Activity (15 min) — Explain to a friend

- Worksheet prompt:

Imagine your Grade 4 cousin asks: “*Why does Nikhilam work? Isn’t it just magic?*” In 4–6 sentences, explain why it’s NOT magic. Use the words **nine**, **ten**, **plus one**, and **place value**.

- 10 min individual writing.
- 5 min pair-share: each partner reads their paragraph; partner says one thing that’s clear and one that’s unclear.

Wrap-up (5 min)

- Read one strong paragraph aloud (with permission).
- Honest historical note: “*Tirthaji’s book (1965) gives this rule. The rule itself is genuinely correct. Tirthaji’s claim that it comes from a Vedic appendix has not been verified by other Sanskritists. We use the rule because it works. The history is younger than the name suggests.*”
- Preview Day 7: “*Tomorrow — a surprise. The same rule is built into how computers subtract. It’s called ‘two’s complement.’*”

Student-facing content

Why Nikhilam works

The key identity:

$$10^n - A = (10^n - 1) - A + 1$$

- $10^n - 1$ is “all nines” (e.g. 9999).
- Subtracting A from “all nines” is trivial digit-by-digit — there’s no borrow because $9 \geq$ any digit.
- The “+ 1” at the end gets absorbed into the last digit, turning “ $9 - \text{last_digit}$ ” into “ $10 - \text{last_digit}$.”

That’s the whole proof.

Tirthaji’s *Vedic Mathematics* (1965) packages this proof into six Sanskrit words: *Nikhilam Navataścaramam Daśataḥ* — “all from nine and the last from ten.” The math is real. The Sanskrit is from a 20th-century synthesis, not a Vedic text.

It is not magic. It is good notation.

Homework

- Find ONE moment in your daily life where you naturally “round to the nearest 100 / 1000” without thinking. Write 2 sentences describing it. (E.g. tipping at a restaurant. Counting change. Comparing prices.)
- Practise: complement of 4321 to 10 000. Show the algebraic identity for this one (the same way the lesson did $1000 - 567$).

Differentiation

- **Tier up:** Write the explanation for a Grade 11 student. Use formal notation. Show the identity holds for any base b : $b^n - A = (b^n - 1) - A + 1$. Preview Day 7’s binary case.
- **Tier down:** Skip the written explanation. Verbalise to your partner. Use the bar-model picture as scaffold.

Teacher notes

- The most common student error in the paragraph: confusing “the last digit takes from 10” with “you add 1 at the end.” These are the SAME operation, just expressed differently — $10 - x = (9 - x) + 1$. Some students will write both; that’s fine.
- Some students prefer the visual bar-model; some prefer the algebra. Both are valid. Honour the choice and ask them to draw the OTHER one as homework.
- Honest provenance is essential here. Don’t oversell. Some students (or parents) believe Nikhilam is “ancient Vedic mathematics.” Clarify gently; preview the Senior-band treatment of the historicity question.

Citation(s) used in this lesson

- Tirthaji, *Vedic Mathematics* (1965), sūtra Nikhilam (#2) — primary source.
- *Microsoft Word - sutras1.pdf* — wall reference for the full 16 sūtra list.

Day 7 — Two’s Complement Connection

Module: Nikhilam Subtraction · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will identify the structural similarity between Nikhilam (“all from 9, last from 10”) in base 10 and **two’s complement** (“flip all bits, add 1”) in base 2, and will subtract two small binary numbers using the two’s-complement method.

Materials

- Whiteboard
- Printed 8-bit binary number line poster (0000 0000 to 1111 1111)
- Pencil + ruled paper
- Optional: a calculator with a binary mode (most smartphone calculators have one)

Vocabulary introduced today

- *binary* — base 2 (the number system computers use).
- *bit* — a binary digit; either 0 or 1.
- *two's complement* — the binary version of the Nikhilam principle.

Lesson flow

Warm-up (5 min)

- Sūtra chant.
- Quick fire: 3 base-10 complements (to 100, 1000, 10 000).
- Provocation: “*Same trick. Different number system. Where else might this work?*”

Core (25 min) — From base 10 to base 2

1. The Nikhilam pattern, restated in base-neutral language:

To compute $b^n - A$:

- Each digit of A : subtract from $(b - 1)$ — the maximum digit in that base.
- The last digit: subtract from b .

In base 10: $(b - 1) = 9$ and $b = 10 \rightarrow$ “all from 9, last from 10.”

2. In base 2. “*The biggest digit in base 2 is 1. So ‘(b - 1)’ is 1. The base b is 2.*”

Pattern becomes:

- Each bit of A : subtract from 1. (This is the same as **flipping the bit**: $0 \rightarrow 1, 1 \rightarrow 0$.)
- The last bit: subtract from 2. ($2 - 0 = 10$ in binary, $2 - 1 = 1$. Equivalent: “flip the last bit and add 1” — except we just add 1 to the whole flipped result and let the carry propagate.)

That’s exactly the **two’s complement** algorithm taught in computer science: > **To negate a binary number: flip all bits, then add 1.**

3. Worked example. Subtract $5 - 3$ using two’s complement, 4 bits:

- $5 = 0101$. $3 = 0011$.
- To compute $5 - 3$, we compute $5 + (-3)$, where -3 is the two’s complement of 3.
- Two’s complement of 3 (4 bits): flip all bits of $0011 \rightarrow 1100$, then add 1 $\rightarrow 1101$.
- Now add: $0101 + 1101 = 1\ 0010$.
- The leading 1 (the 5th bit) is the **overflow / carry out** — we discard it. (This carry-out is the “underflow” of the $16 = 2^4$ base, exactly like the $10\ 000$ in $10\ 000 - 4567$.)
- Result: $0010 = 2$. ✓

4. Why this is the SAME as Nikhilam.

- In base 10: $10\ 000 - 4567 = 5433$. Same calculation: complement of 4567 = 5433, and $4567 + 5433 = 10\ 000$.
- In base 2: $16 - 3 = 13 = 1101$. That's the two's complement of 3. And $0011 + 1101 = 1\ 0000 = 16$. ✓
- Then we subtract THAT result from 5 (or equivalently add it to 5 modulo 16): $5 - 3 \equiv 5 + (16 - 3) \pmod{16} = 5 + 13 = 18 \equiv 2 \pmod{16}$.

“Tirthaji’s six-word sūtra and the two’s-complement circuit in your phone’s CPU are doing the same algebra. The CPU just does it in base 2.”

5. **Why computers use this.** A CPU has no separate “subtract” circuit — it has an “add” circuit, an “invert bits” circuit, and a “+1” (increment) circuit. Two’s complement lets it do subtraction with just those. This is the deepest reason: the Nikhilam principle saves silicon.

Activity (10 min) — Try it

- Worksheet: 4 small subtractions to do via two’s complement (4 bits, all results positive).

Problem	A	B	Flip A or B?	Two’s-comp	Add	Discard top bit	Answer
6 – 4	0	0	Flip B	1100	1	drop top → 0010	2 ✓
	1	1			001		
	1	0			0		
	0	0					
7 – 5	0	0	Flip B	1011	1	drop top → 0010	2 ✓
	1	1			001		
	1	0			0		
	1	1					
9 – 2	1	0	Flip B	1110	1	drop top → 0111	7 ✓
	0	0			011		
	0	1			1		
	1	0					
12 – 7	1	0	Flip B	1001	1	drop top → 0101	5 ✓
	1	1			010		
	0	1			1		
	0	1					

- Students work in pairs. One does the binary math; the other verifies with a calculator.

Wrap-up (5 min)

- “You’ve just done in your head what a CPU does in nanoseconds. Different scale, same logic.”
- Preview Day 8: “Tomorrow — what if the number we’re subtracting FROM isn’t a clean power of ten? E.g. $8243 - 567$. Nikhilam still helps, but we need one more move.”

Student-facing content

Two’s Complement = Nikhilam in Binary

	Base 10 (Nikhilam)	Base 2 (Two's Complement)
Maximum digit	9	1
“All from (max)”	All from 9	Flip each bit
“Last from base”	Last from 10	Add 1

Same algebra. Same trick. The CPU inside your phone uses this every time it computes a subtraction.

“*Nikhilam Navataścaramam Daśataḥ*” — Tirthaji’s 1965 sūtra. The base-2 cousin was independently discovered in 1945 by the team designing the EDVAC computer.

Homework

- Convert $13 - 5$ to binary (4 bits) and solve via two’s complement. Show your work.
- One reflection sentence: “*What surprised me about the binary connection.*”

Differentiation

- **Tier up:** Try **negative results**. $3 - 5$ in 4 bits gives 1110, which is the two’s-complement representation of -2 . Look up how CPUs read this.
- **Tier down:** Skip the binary math. Just understand the structural parallel: “flip the digits, add 1, same idea.” Repeat the base-10 examples instead.

Teacher notes

- This is the most cross-disciplinary day of the module. Some students will be enthralled; others will find binary intimidating. Both reactions are fine.
- Many students have *heard* of binary; few have *done* arithmetic in it. The 4-bit examples are deliberately small.
- A natural question: “*Does this mean Tirthaji invented two’s complement?*” — No. Two’s complement was independently developed for digital computing in the 1940s. What’s beautiful is the **structural parallel** — the same algebraic principle ($b^n - A = (b^n - 1) - A + 1$) plays the same role in both base 10 and base 2.

Citation(s) used in this lesson

- Tirthaji, *Vedic Mathematics* (1965), sūtra Nikhilam (#2).
- Patterson & Hennessy, *Computer Organization and Design*, two’s-complement section — standard CS reference for the binary parallel.

Day 8 — General Subtraction with Complements

Module: Nikhilam Subtraction · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will subtract a number from any larger number (not necessarily a power of ten) using a *combined* method: take the complement of the subtrahend to a nearby base, add it to the minuend, and adjust by the difference.

Materials

- Whiteboard
- activities/activity-02-shopkeeper.md worksheet
- Pretend currency (paper notes of 100, 500, 1000 rupee denominations)
- Stopwatches

Lesson flow

Warm-up (5 min)

- Sūtra chant.
- Quick fire: 4 complements (mix of bases).

Core (15 min) — Beyond clean bases

1. **The setup.** “So far we’ve done $1000 - 567$. Easy: complement of 567 is the answer, 433. Now: $8243 - 567$. The minuend isn’t a clean power of ten. What do we do?”

2. **Idea: borrow a base.** Rewrite:

$$\begin{aligned}8243 - 567 \\&= 8243 - 1000 + (1000 - 567) \\&= 7243 + 433 \\&= 7676.\end{aligned}$$

“We pick a base that’s larger than the subtrahend ($1000 > 567$ here). We subtract that base from the minuend ($8243 - 1000 = 7243$; easy). Then we add the complement (433). Done.”

Verify: $8243 - 567 = 7676$ by standard borrowing. ✓

3. **Why does this work?** Algebra:

$$M - S = M - (B - (B - S)) = M - B + \text{complement}(S \text{ to } B)$$

“ $B - S$ is the complement of S to base B . Add the complement and subtract B from M .”

4. **Worked examples (class participates):**

Problem	Base B	Complement	$M - B$	Add	Answer
$8243 - 567$	1000	433	7243	$7243 + 433$	7676
$5172 - 89$	100	11	5072	$5072 + 11$	5083

Problem	Base B	Complement	$M - B$	Add	Answer
$6300 - 248$	1000	752	5300	$5300 + 752$	6052
$12\ 000 - 4\ 567$	10 000	5 433	2 000	$2\ 000 + 5\ 433$	7 433

5. **When is this faster than standard borrowing?** “When the subtrahend is close to a base — say 567 to 1000 — Nikhilam-with-adjustment is genuinely faster. When the subtrahend is in the middle (like 4567 from 8243, where 4567 is not close to any clean base), standard borrowing is competitive. The mature mental-arithmetic move is to know both and pick.”

Activity (20 min) — Shopkeeper role-play

See activities/activity-02-shopkeeper.md.

- In pairs, one is the shopkeeper, one is the customer.
- Customer hands over a “round” note (100, 500, 1000, or 2000 rupees).
- Shopkeeper computes the change using Nikhilam (mental).
- Swap roles every 3 rounds.
- 10 scenarios per pair. Times recorded.

Wrap-up (5 min)

- Quick discussion: “Where in real life does this method save time?” Most answers will be shop-related. Affirm and add: change-counting, expense splitting, deficit tracking.
- Preview Day 9: “Tomorrow — speed test, both methods, head-to-head.”

Student-facing content

General Subtraction with Complements

To compute $M - S$:

1. Pick a base B such that $B > S$ and B is a power of 10 (so 100, 1000, 10 000, ...).
2. Compute the complement of S to B (using Nikhilam) — call it C .
3. Compute $M - B$ (which is easy, since B is a round number).
4. Answer = $(M - B) + C$.

Algebraic identity: $M - S = (M - B) + (B - S)$

Verify by checking: $M - S$ should equal your answer.

Example: $8243 - 567 = (8243 - 1000) + 433 = 7243 + 433 = \mathbf{7676}$.

Homework

- Solve 6 mixed subtractions using the combined method:
 - $7234 - 587$ (base 1000)
 - $9\ 612 - 3\ 458$ (base 10 000)
 - $5\ 100 - 67$ (base 100)

- $13\,405 - 2\,850$ (base 10 000)
- $800 - 39$ (base 100)
- $25\,000 - 4\,999$ (base 10 000)
- For each, show the base, the complement, and the final answer.

Differentiation

- **Tier up:** What if the subtrahend has *more* digits than the base needs? E.g. $8243 - 5567$ — using base 10 000, complement of 5567 is 4433, and $8243 - 10\,000 = -1757$, so answer = $-1757 + 4433 = 2676$. Verify: $8243 - 5567 = 2676$. ✓. Negative intermediate results are fine.
- **Tier down:** Use base 100 for everything. Only graduate to 1000 when comfortable.

Teacher notes

- The combined method introduces a real subtlety: students must now choose a base. Some over-pick (use 10 000 for $5172 - 89$ when 100 would do). Some under-pick (use 100 for $8243 - 567$). Either causes errors. Teach: “pick the smallest base that still exceeds the subtrahend.”
- The shopkeeper role-play tends to make this concrete. Make sure the “round notes” are physically present — the mental model “I have 1000 in my hand, I need to give 567 of it” makes the algebra intuitive.
- Some classes will benefit from spending the whole lesson on the shopkeeper drill and pushing the algebra to Day 9. That’s a fine reordering.

Citation(s) used in this lesson

- Tirthaji, *Vedic Mathematics* (1965), sūtra Nikhilam (#2) — primary source.
- *FUNDAMENTAL AND VEDIC MATHEMATICS .pdf*, Nikhilam chapter — provided the general-case worked examples that informed today’s table.

Day 9 — Speed Practice + Method Comparison

Module: Nikhilam Subtraction · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will solve a mixed set of subtraction problems using both standard borrowing and Nikhilam, will compare the two methods on each problem, and will articulate a personal heuristic for when to pick which method.

Materials

- Comparison worksheet (10 problems, two-column answer space — one column per method)
- Stopwatches
- Whiteboard

- Final project work-time at the end

Lesson flow

Warm-up (5 min)

- Sūtra chant.
- Quick fire: 4 complements (mixed bases).
- Acknowledge: tomorrow is the summative.

Speed test (15 min) — head-to-head

Distribute the comparison worksheet. 10 problems, ordered roughly:

#	Problem	Likely faster method	Why
1	$100 - 37$	Nikhilam	Pure power-of-ten subtraction
2	$87 - 24$	Standard	Small, no borrow either way
3	$1000 - 463$	Nikhilam	Power-of-ten
4	$532 - 178$	Standard	No clean base nearby
5	$10\,000 - 5827$	Nikhilam	Power-of-ten
6	$7000 - 248$	Nikhilam (with adjust)	Round multiple of 1000
7	$9385 - 4716$	Tie / standard	Mid-range subtrahend
8	$50\,000 - 12\,345$	Nikhilam (with adjust)	Round multiple of 10 000
9	$8243 - 567$	Nikhilam (with adjust)	Subtrahend close to 1000
10	$4321 - 4118$	Standard	Both

#	Problem	Likely faster method	Why
			small, similar magnitudes

Students solve each problem **once** by their chosen method, then again by the other method. Time both. Mark which method was faster.

Discussion (10 min) — When does each win?

Compile on board: which problems were faster with Nikhilam? Which with standard?

Heuristic that students will surface:

- **Nikhilam wins** when the minuend is a clean power of ten OR the subtrahend is close to a clean power of ten.
- **Standard wins** when both numbers are mid-range and the subtraction requires only one or two borrows.
- **Tie** when the numbers are small.

Write the heuristic on the board. Encourage students to copy it into their workbook.

Honest note: *“This is a real piece of mathematical maturity — knowing when to use which tool. Speed comes from knowing the choice, not from forcing one method everywhere.”*

Project work (12 min)

- Last in-class time before the showcase. Pairs work on their project artefact.
- Teacher circulates. Check: does each project have at least one Nikhilam-method example? Does each cite a source?

Wrap-up (3 min)

- Tomorrow’s plan: 20-min summative quiz, then 3-min project presentations.
- Sūtra chant — final practice for tomorrow’s recitation.

Student-facing content

When to use Nikhilam vs. Standard Borrowing

Situation	Faster method
Subtracting from 100, 1000, 10 000, ...	Nikhilam
Subtracting a small number from a round multiple of 100/1000	Nikhilam with adjust
Both numbers in the middle, similar sizes	Standard
Both numbers small (under 100)	Either

The mature move: know both methods; pick the right one for the problem.

Homework

- Final project touches.
- 5 minutes of sūtra recitation practice — say it 5 times before bed.
- Bring all project materials tomorrow.

Differentiation

- **Tier up:** Mental challenge — do 10 mixed problems in your head with no paper. Time goal: under 3 minutes.
- **Tier down:** Worksheet only. Don't worry about choosing methods — solve each by Nikhilam where you can, standard otherwise.

Teacher notes

- The honest framing of “both methods are tools” matters. Avoid letting students walk away thinking “Nikhilam is always faster.” It isn't. It's faster for *certain shapes* of problem.
- Some students will discover the heuristic naturally. Highlight their wording — let them name the rule.
- Project check is critical today. If any pair is behind, give them homework instructions explicitly.

Citation(s) used in this lesson

- Tirthaji, *Vedic Mathematics* (1965), sūtra Nikhilam (#2) — primary source.

Day 10 — Final Assessment + Project Share

Module: Nikhilam Subtraction · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will complete the summative quiz, present their project artefact in 3 minutes, and submit a brief written self-reflection.

Materials

- quizzes/summative.md (printed, one per student)
- Stopwatch for project presentations
- Camera (phone) for documenting projects, if permitted by school policy
- Stickers for completion (optional, but motivating)

Lesson flow

Warm-up (3 min)

- Final sūtra recitation, in unison, from memory — *Nikhilam Navataścaramam Daśataḥ*.

- One sentence each, going around the room: “*The one thing I’ll remember from this module is...*”

Summative quiz (20 min)

- Closed notebook.
- 25 marks. Sections: complement recall, subtraction practice, general subtraction, conceptual short answer, open response on “why it works.” See `quizzes/summative.md`.
- Strict 15-min time limit on the multiple-choice / short-answer parts; remaining time for the open response.
- Collect.

Project presentations (18 min)

- Each pair presents for 3 minutes (set the timer visibly).
- Format depends on artefact:
 - **Shopkeeper chart:** Pair demos using pretend currency on a real scenario.
 - **Teaching video:** Pair plays the 60-sec video, then takes 90 sec of Q&A.
 - **Two’s-complement poster:** Pair walks through the poster’s argument.
- Rest of class fills in a quick “feedback card” — one strength, one question.

Wrap-up (4 min)

- Quick poll: was Module 9 useful? Surprising? Hard?
- Honest history note one more time: “*You learned a real, useful method. The sūtra is from Tirthaji’s 1965 book. The deeper history is more recent than the name ‘Vedic’ suggests. You can hold both: the method is real; the historical claim is unverified.*”
- Self-reflection slip (1 minute): on the back of the quiz, complete:
 - One thing I learned: _____
 - One thing I’m still confused about: _____
 - My favourite Sanskrit term from the module: _____

Student-facing content

Today’s tasks

1. Summative quiz (20 min, closed notebook).
2. Project presentation (3 min, with your partner).
3. Self-reflection (1 min, on the back of the quiz).

You’ve come a long way. Day 1’s 10 000 – 4567 took most of you about 30 seconds. By now you can do it in 5.

Homework

- None. Module ends.

Differentiation

- **Tier up:** If finished early on the quiz, write the algebraic identity for the Nikhilam method in full generality (for any base b).
- **Tier down:** Extra 5 min on the quiz allowed for slow processors; quiet pull-aside support.

Teacher notes

- The summative is the most marked single assessment of the module. Be strict on closed-notebook BUT generous on the open-response question — student paraphrases of the algebraic identity are valid even if not perfectly phrased.
- Project presentations: keep the energy up. Use stickers / acknowledgment for every pair. The presentation is partly a performance; affirm the work.
- If a pair didn't finish their artefact, still let them present. The presentation itself is graded; the artefact is graded separately on the rubric.
- File the quiz results AND the project rubric scores to inform pacing for next year's cohort.

Citation(s) used in this lesson

- Tirthaji, *Vedic Mathematics* (1965), sūtra Nikhilam (#2).

Module 9 (Middle) — Formative Quizzes

Two parts: Part A on Day 3, Part B on Day 5.

Part A — Day 3 (5 min, 5 short problems)

Name: _____ **Class:** _____

Find the complement of each number to the next power of ten. Use the Nikhilam rule: all from 9, the last from 10.

1. Complement of **47** to 100: _____

2. Complement of **283** to 1000: _____

3. Complement of **904** to 1000: _____

4. Complement of **3 752** to 10 000: _____

5. State the Nikhilam sūtra (in English is fine):

Part A answer key

Q	Answer	Common error
1	53 (9-4=5, 10-7=3)	Mixing 9 and 10 columns
2	717 (9-2, 9-8, 10-3)	Last digit “from 9” instead of “from 10”
3	96 — i.e. 096 (9-9, 9-0, 10-4)	Forgetting the leading zero
4	6 248 (9-3, 9-7, 9-5, 10-2)	Off-by-one on a digit
5	“All from nine and the last from ten” (or the Sanskrit <i>Nikhilam Navataścaramam Daśataḥ</i>)	Vague paraphrase

Part B — Day 5 (12 min, 12 mixed problems)

Name: _____ Class: _____

Section 1 — Complement recall (3 marks)

1a. Complement of **62** to 100: _____ 1b. Complement of **148** to 1000: _____ 1c. Complement of **4 591** to 10 000: _____

Section 2 — Power-of-ten subtractions (4 marks)

2a. $1000 - 358 =$ _____ 2b. $10\ 000 - 6\ 427 =$ _____ 2c. $1000 - 72 =$ _____ (Hint: pad with leading zero.) 2d. $100\ 000 - 12\ 345 =$ _____

Section 3 — Padding cases (3 marks)

3a. $1000 - 9 =$ _____ 3b. $10\ 000 - 87 =$ _____ 3c. $1000 - 999 =$ _____

Section 4 — Short answer (2 marks)

4. Why does the LAST digit take “from 10” instead of “from 9”? Answer in one sentence.

Part B answer key

Q	Answer	Marks
1a	38	1
1b	852	1
1c	5 409	1
2a	642	1
2b	3 573	1
2c	928	1
2d	87 655	1

Q	Answer	Marks
3a	991 (pad 9 → 009)	1
3b	9 913 (pad 87 → 0087)	1
3c	001 = 1	1
4	“Because $(10 - x) = (9 - x) + 1$, and that ‘+1’ is the carry from $10^n - 1$ to 10^n .” (Variants acceptable that mention the +1 needing to be absorbed somewhere.)	2 (full = mentions both the +1 and where it goes; 1 = mentions only one)

Total Part B: 12 marks.

What to do with the results

- Class average below 7/12 → reteach Day 6 on “why it works” before moving to Day 7. Reinforce padding with quick warm-up drills.
- Individual below 4/12 → pull aside on Day 7 for 10-min reteaching of complements + padding.
- Section 3 (padding) is the highest-error section in field-test data. Don’t be surprised if it pulls scores down. Address explicitly.

Module 9 (Middle) — Summative Quiz

Time: 20 min · **Marks:** 25 · **Closed notebook**

Instructions: Name + class at top. No calculators. Show working where space allows.

Name: _____ **Class:** _____

Section A — Sūtra Recall (2 marks)

1. Write the Nikhilam sūtra in (a) Sanskrit (IAST or transliteration accepted) and (b) English.

Section B — Complement Recall ($4 \times 1 = 4$ marks)

Find the complement to the indicated base.

2. **45** to 100: _____
3. **372** to 1000: _____
4. **803** to 1000: _____
5. **6 528** to 10 000: _____

Section C — Power-of-Ten Subtractions ($4 \times 1.5 = 6$ marks)

Use Nikhilam. Show your digit-by-digit working.

6. $1000 - 567 =$ _____

7. $10\ 000 - 3\ 478 =$ _____

8. $100\ 000 - 12\ 345 =$ _____

9. $10\ 000 - 89 =$ _____ (Don't forget to pad!)

Section D — General Subtraction ($2 \times 2 = 4$ marks)

Use the combined method (complement + base-adjust). Show your working.

10. $8\ 243 - 567 =$ _____

11. $25\ 000 - 4\ 999 =$ _____

Section E — Binary Bonus (2 marks)

12. Convert these single-digit binary numbers and compute via two's complement (4 bits):

$7 - 5 = ?$

Working: $7 =$ _____, $5 =$ _____, two's-complement of $5 =$ _____, sum = _____, drop top bit = _____. Answer: _____

Section F — Open Response (5 marks)

13. A Grade 4 student asks: “*Why does the Nikhilam method work? Isn't it just magic?*” In 4–6 sentences, explain why it's NOT magic. Use the words **nine**, **ten**, **plus one**, and **place value**.

Section G — Honesty Question (2 marks)

14. Tirthaji's *Vedic Mathematics* (1965) claims the Nikhilaṃ sūtra comes from a Vedic appendix-text. Is this claim well-supported? Answer in one sentence, then say what is definitely true.

Well-supported? (Y / N): _____

What is definitely true: _____

Answer key

Q	Answer	Marks
1a	<i>Nikhilaṃ Navataścaramaṃ Daśataḥ</i> (close transliteration acceptable)	1
1b	“All from nine and the last from ten”	1
2	55	1
3	628	1
4	197	1
5	3 472	1
6	433 (working: 9-5, 9-6, 10-7)	1.5
7	6 522 (9-3, 9-4, 9-7, 10-8)	1.5
8	87 655 (9-1, 9-2, 9-3, 9-4, 10-5)	1.5
9	9 911 (pad 89 → 0089; 9-0, 9-0, 9-8, 10-9)	1.5
10	7 676 (base 1000; complement 433; 8243–1000+433=7676)	2
11	20 001 (base 10 000; complement 5 001; 25 000–10 000+5 001=20 001). <i>Alt accepted: 5001 mistake → -1 if marked otherwise correct.</i>	2
12	7 = 0111, 5 = 0101, two's-comp of 5 = 1011, 0111+1011 = 1 0010, drop top → 0010 = 2	2
13	Open response: full = mentions 9, 10, +1, and place value, with a correct logical flow. 4 = three of the four key words / correct logic. 3 = two key words / mostly correct. 2 = one key word /	5

Q	Answer	Marks
	partially correct. 1 = attempts.	
14	N. Definitely true: the method works; Tirthaji synthesised the 16 sūtras; the deeper Vedic provenance is unverified.	2 (1 for the N; 1 for the “definitely true” sentence)

Total: 25 marks.

Grading bands (project combined → out of 40): - 32–40: Exceeds - 24–31: Meets - 16–23: Approaches - 0–15: Beginning

Homework

Homework — Module 9 Nikhilam Subtraction (Middle)

Day 1 → Day 2

Task: Memorise the sūtra. Practice 10 complements to 100: $73 \rightarrow ?$, $41 \rightarrow ?$, $28 \rightarrow ?$, $86 \rightarrow ?$, $95 \rightarrow ?$, $17 \rightarrow ?$, $60 \rightarrow ?$, $52 \rightarrow ?$, $34 \rightarrow ?$, $89 \rightarrow ?$ Time goal: under 30 seconds total.

Time: 5 min.

Day 2 → Day 3

Task: Have a parent or sibling call out 20 two-digit numbers in 90 seconds; you say the complement to 100 for each. Track how many you got right.

Time: 5 min.

Day 3 → Day 4

Task: Find the complement to 1000 of each: 156, 423, 789, 802, 64 ($\rightarrow 064$), 7 ($\rightarrow 007$), 555, 900. Find the complement to 10 000 of each: 3 456, 1 027, 8 888, 75 ($\rightarrow 0075$), 999. Total time goal: under 2 minutes.

Time: 10 min.

Day 4 → Day 5

Task: 12 power-of-ten subtractions on the back of today’s worksheet. Self-check by adding (number + your answer should equal the base).

Time: 15 min.

Day 5 → Day 6

Task: Submit your project choice + a 3-sentence plan. (One of: shopkeeper chart / teaching video / two's-complement poster.) Plus: 5 mixed subtractions — $1000 - 358$, $10\ 000 - 6\ 427$, $1000 - 72$, $100\ 000 - 12\ 345$, $10\ 000 - 9$.

Time: 15 min.

Day 6 → Day 7

Task: Find ONE moment in your daily life where you naturally “round to the nearest 100 / 1000” without thinking. Write 2 sentences. Example: rounding a restaurant tip, counting change, comparing prices.

Plus: redo today's algebra example with $10\ 000 - 4321$. Show the identity $10^n - A = (10^n - 1) - A + 1$ for this case.

Time: 10 min.

Day 7 → Day 8

Task: Convert $13 - 5$ to binary (4 bits) and solve via two's complement. Show your work. Plus: one reflection sentence — what surprised you about the binary connection.

Time: 10 min.

Day 8 → Day 9

Task: 6 mixed general subtractions: $-7\ 234 - 587$ (base 1000) - $9\ 612 - 3\ 458$ (base 10 000) - $5\ 100 - 67$ (base 100) - $13\ 405 - 2\ 850$ (base 10 000) - $800 - 39$ (base 100) - $25\ 000 - 4\ 999$ (base 10 000)

For each, show your chosen base, the complement, and the answer.

Time: 20 min.

Day 9 → Day 10

Task: Final project touches. 5 minutes of sūtra recitation. Bring all project materials. Be ready to present in 3 minutes.

Time: 15 min plus material gathering.

Writing Prompts

Writing Prompts — Module 9 Nikhilam Subtraction (Middle)

Eight prompts. Useful as exit tickets, homework reflections, or project warm-ups.

Reflection prompts

1. (Day 1, end of class, 3 min) Describe what it felt like to subtract $10\ 000 - 4567$ under time pressure. Did your mind borrow? Did you count? Did you panic? Be honest — there's no wrong answer.

2. (Day 5, after the retest, 5 min) You re-solved the same problem with Nikhilam. Did your time change? Did the *feeling* of doing the math change? Compare the two experiences in your own words.

Explanatory prompts

3. (Day 3 or Day 6, homework, ~120 words) Explain Nikhilam to a Grade 4 student who has never seen it. No more than 6 sentences. Use one specific example with verification.

4. (Day 7, homework, ~120 words) Explain in your own words why the same idea works in base 2 (binary) as in base 10. Use one binary example. Don't use the phrase "two's complement" without explaining what it means.

Argumentative prompt

5. (Day 6 or homework, 1 paragraph) Some people claim the Nikhilam sūtra is "ancient Vedic mathematics." Others say it's a 20th-century synthesis published by Tirthaji in 1965. Pick a position and defend it in 5–7 sentences. Reference what you know.

Creative prompts

6. (Day 4 or Day 5, optional, 1 page) Write a short script (50–150 words) in which a frustrated student and a confident Nikhilam user have a conversation. The frustrated student raises one good objection to the method. The confident user gives an honest answer. (No "and then they magically agreed" endings.)

7. (Day 7 or Day 8, optional, 1 page) Imagine the CPU inside your phone could talk. It explains how it just subtracted $7 - 3$. Write a 100–200 word monologue from the CPU's point of view, in everyday language, but accurate about what it actually does.

Cross-disciplinary prompt

8. (Day 7, optional, 1 paragraph) This module touched mathematics, computer science, and Sanskrit. In 3–5 sentences, describe one *other* place in your life or studies where the same idea (rounding up to a base, adjusting by the difference) shows up.

Prompt-to-band notes

- For **Primary** band, prompts 1, 3, and 6 work as drawings or oral responses.
- For **Senior** band, prompt 5 deserves a full essay treatment with citations from sources .md. Prompt 4 can be extended into a base- n general analysis.

How to give feedback

- Prompts 1, 2: acknowledge with one sentence — no correction. Listening, not judging.
- Prompts 3, 4: respond with “the part that landed” and “the part that confused me.”
- Prompt 5: use the rubric — does the writing acknowledge the other side, give a specific example, reach a clear position?
- Prompts 6, 7, 8: no grading. Read the best ones aloud (with permission).

Activity 01 — Complement & Subtraction Drill

Module: Nikhilam Subtraction · **Band:** Middle · **Day:** 4 · **Time:** 20 min

Type

Paired practice with timed self-correction.

Goal

Each student solves 8 power-of-ten subtraction problems using Nikhilam, with the partner checking accuracy.

Materials per pair

- 1 printed problem sheet (8 problems, see below)
- 1 stopwatch
- 1 pen each
- Pocket calculator (only for the verification round)

Set-up (teacher, before class)

- Print the problem sheet. You can vary numbers — keep difficulty graded from $1000 - N$ to $100 - N$.

Instructions (student-facing handout)

Complement & Subtraction Drill — work in pairs.

1. Both partners solve all 8 problems individually (8 min). Use the Nikhilam method. Don't look at each other's work.
2. Swap sheets. Partner-A reads problem 1; partner-B re-solves on a fresh sheet (no peeking at A's answer). Compare. If they differ, work out which is correct.
3. Repeat for problems 2–8.
4. Last 4 min: verify any 2 problems using a calculator (number + your complement should equal the base; or compute the subtraction directly).

Problem sheet

1. $1000 - 427 = ?$
2. $1000 - 892 = ?$
3. $1000 - 65 = ?$ (pad: 065)
4. $10000 - 3478 = ?$
5. $10000 - 1294 = ?$
6. $10000 - 87 = ?$ (pad: 0087)
7. $100000 - 23\ 456 = ?$
8. $100000 - 7\ 089 = ?$ (pad: 07 089)

Answers (teacher copy):

- | | | | |
|---------|---------|-----------|-----------|
| 1. 573 | 2. 108 | 3. 935 | 4. 6522 |
| 5. 8706 | 6. 9913 | 7. 76 544 | 8. 92 911 |

Discussion prompts (last 2 min)

- Which problem felt fastest? Why? (Likely #1–3, smaller bases.)
- Which one tripped you up? (Likely #3 or #6 — the padding cases.)
- Did your partner solve any differently than you?

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Pair compares specific digits; debates the padding step	"I think 65 needs to be 065 for this to work"
Needs intervention	One partner just copies; long silences	"Whatever, sure"

Safety / sensitivity

- No public timed competition between pairs. Pairs are self-pacing.

Differentiation

- **Tier up:** Mixed sheet — half power-of-ten subtractions, half “general” subtractions with base-adjustment (preview Day 8).
- **Tier down:** All 8 problems are $1000 - N$. Build fluency at one base before moving up.

Clean-up

- Worksheets collected for teacher spot-check before Day 5.

Activity 1 — Complement Flashcards

Module: Subtraction — Nikhilam (All from 9) · **Band:** Middle · **Duration:** 20 min

Purpose

Reinforce the day-by-day concepts of Subtraction — Nikhilam (All from 9) through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

Materials

40 flashcards (target – number) pre-made

What students do

In pairs, one student holds up “1000 – 738” (etc.), the other shouts the answer within 3 seconds using the Nikhilam method.

Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.
2. **Working phase (10 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

Safety

- General safety briefing before any materials are handled.
- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

Activity 02 — Shopkeeper Role-Play

Module: Nikhilam Subtraction · **Band:** Middle · **Day:** 8 · **Time:** 20 min

Type

Paired role-play with mental computation.

Goal

Each pair runs through 10 “shop” transactions, computing change using Nikhilam. The customer hands a round-denomination note; the shopkeeper returns the change.

Materials per pair

- A small stack of pretend currency: 100s, 500s, 1000s, 2000s (3–4 of each)
- 1 printed scenario sheet (10 items with prices, see below)
- 1 stopwatch
- Whiteboard / scrap paper (allowed but discouraged — push for mental)

Set-up (teacher, before class)

- Print the scenario sheet (vary the items if you like — keep prices in the 50–950 range so all transactions involve a clean base).
- Distribute pretend currency.

Instructions (student-facing handout)

Shopkeeper Role-Play — work in pairs.

1. Decide who’s the shopkeeper, who’s the customer. Swap after 5 rounds.
2. Customer reads the item + price from the sheet (“One book for 367 rupees”).
3. Customer hands over a round note (100, 500, 1000, or 2000) that’s larger than the price.
4. **Shopkeeper computes change mentally using Nikhilam.** Says the amount aloud. Hands over corresponding currency.
5. Customer verifies. Mark ✓ or ✗ on the sheet.
6. Time the whole round: 60-second target per transaction.

Scenario sheet

#	Item	Price (₹)	Customer pays (₹)	Change (₹)	Shopkeeper’s working
1	Note book	73	100	_____	_____
2	Pen	28	100	_____	_____
3	Lunch	367	500	_____	_____
4	Book	482	1000	_____	_____
5	Toy	199	500	_____	_____
6	Shoes	1450	2000	_____	_____
7	Box	235	1000	_____	_____

#	Item	Price (₹)	Customer pays (₹)	Change (₹)	Shopkeeper's working
	of sweet s				
8	Back pack	875	1000	_____	_____
9	T- shirt	549	1000	_____	_____
1 0	Movi e ticket s (2)	760	2000	_____	_____

Answers (teacher copy):

1. 27	2. 72	3. 133	4. 518
5. 301	6. 550	7. 765	8. 125
9. 451	10. 1 240		

Discussion prompts (last 2 min)

- Which transaction felt the slowest? Why? (Likely #6 or #10 — the 2000-rupee ones; less practised base.)
- Did you give correct change every time? If not, where did the error creep in?
- A real shopkeeper does this all day. How long do you think it took them to get fluent?

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Quick mental call; both partners watching the math	“133. Done.”
Needs intervention	Long pause; reaching for paper every time	“Wait... hmm...”

Safety / sensitivity

- Money themes can be sensitive in some classrooms. Pretend currency is fine; avoid framing as “the rich shopkeeper / the poor customer.” Keep it neutral.

Differentiation

- **Tier up:** Customer hands a NON-round note (e.g. 870 for a 367-rupee item). Shopkeeper computes $870 - 367 = 503$. This requires the Day-8 general method.
- **Tier down:** Use only 100-rupee notes for all transactions. Build fluency at base 100.

Clean-up

- Currency back in the teacher's box.
- Scenario sheets collected for teacher review.

Extension: project hook

- Pairs whose project is the “shopkeeper’s quick-change chart” use today’s activity as direct material. Their chart should help a real shopkeeper give change for at least three round-denomination notes.

Project Brief

Project Brief — Module 9 Nikhilam Subtraction (Middle)

Work sessions: Day 8 + Day 9 (in-class time) · **Presentation:** Day 10 · **Total marks:** 40 (project 30 + presentation 10; honesty bonus up to +2)

The project

Each pair (or individual, with teacher approval) chooses ONE of three formats:

Format 1: Shopkeeper’s Quick-Change Chart

A one-page reference card a real shopkeeper could use to give change instantly from 100, 500, 1000, or 2000 rupee notes.

Constraints: - One A4 page (front and back allowed). - Must cover at least 3 round-note denominations. - For each, show the Nikhilam method at least once worked out fully. - Title block with cited source: *Tirthaji, Vedic Mathematics (1965), sūtra Nikhilam*. - Bonus points if the chart is printable AND given to a real shopkeeper for feedback (collect the feedback as a sentence at the bottom).

Format 2: 60-second Teaching Video

A short video that teaches the Nikhilam method to a Grade 4 student.

Constraints: - Total length: 60 seconds (± 10 sec). - Must show: the Sanskrit sūtra, at least one worked subtraction (with the answer verified by addition). - Must cite the source (PDF / book + sūtra name) in the title or closing card. - Sound: must be audible.

Format 3: Two’s-Complement Explainer Poster

A poster connecting Nikhilam (base 10) to two’s complement (base 2). Most challenging.

Constraints: - One A2 or larger sheet (or digital equivalent). - Side-by-side comparison: base 10 and base 2. - Worked example in each base. - Title with cited sources: Tirthaji + one CS textbook (Patterson & Hennessy is fine, or your school’s CS book). - A clear final sentence answering: “Why does the same idea work in both bases?”

Why this matters

You’re moving from *learner* to *teacher* (or *engineer* in the poster case). Each role is harder than just doing the problems. Explaining or applying forces you to understand more deeply.

What success looks like

A strong project will: - Demonstrate the Nikhilam method correctly (no arithmetic errors). - Be pitched for the intended audience (a shopkeeper / a 9-year-old / a CS-curious peer). - Show the pair's own thinking (a unique scenario, a custom example). - Be clearly communicated.

Working in pairs

- Pair work strongly encouraged.
- Individual allowed for the chart or video format.
- Groups of 3 acceptable — each member must have a distinct role.

Citation rules

- At minimum **one** citation must appear in your project — visible.
- Format: *Tirthaji, Vedic Mathematics (1965), sūtra Nikhilam*.
- See `sources.md` for the full list.

Honesty bonus (up to +2)

You can earn up to 2 bonus marks for explicitly addressing the historicity question — i.e., correctly stating that the method is in Tirthaji (1965) and that the deeper “Vedic” provenance is unverified. We reward this because it models the intellectual stance we want.

Self-reflection (submitted with project on Day 10)

In 4–6 sentences: 1. What did you set out to make, and what did you actually end up with? 2. What's the most surprising thing you discovered while making this? 3. If you had three more days, what would you do differently?

Logistics

- Day 5: project choice submitted.
- Day 8: substantive draft / first artefact.
- Day 9: revisions + presentation rehearsal.
- Day 10: 3-min presentation + submission.

Rubric

Project Rubric — Module 9 Nikhilam Subtraction (Middle)

Project marks: 30 (out of 40 with presentation 10).

Criteria

Criterion	4 — Exceeds	3 — Meets	2 — Approaches	1 — Beginning	Weight
Method demonstrated correctly	Method shown completely (digit-wise complement, padding, verification); at least 3 worked examples with no errors.	Method shown completely; minor errors on 1 example.	Method shown but with a conceptual gap (e.g. forgetting the “last from 10” rule).	Method incorrect or absent.	× 2
Use of sources	Cites Tirthaji (1965) AND one supporting source (PDF / textbook) correctly.	Cites at least one source correctly.	Mentions a source vaguely.	No source cited.	× 1.5
Audience suitability	Pitched precisely to the target audience (Grade 4 for video; real shopkeeper for chart; CS-curious peer for poster).	Mostly suitable; minor tonal issues.	Suitable in places, off in others.	Inaccessible for target audience.	× 1.5
Originality	The project shows the pair’s own thinking — a unique scenario, a custom narrative, or a new framing.	Clearly the pair’s own work.	Largely repeated from class examples.	Copied verbatim from a source.	× 1
Communication / craft	Visuals, layout, narration are clear, deliberate,	Communicates clearly. 1–2 craft issues.	Message comes through with effort.	Hard to follow.	× 1

Criterion	4 — Exceeds	3 — Meets	2 — Approaches	1 — Beginning	Weight
	polished. No arithmetic errors shown.				

Maximum: 28. Project score = (sum × weights) × 30 / 28.

Presentation (10 marks)

Criterion	4	3	2	1	Weight
Within time (3 min)	2:45–3:00	2:30–3:15	2:15–3:30	Outside	× 1
Both partners contribute	Equal	Both speak unequally	One dominates	One silent	× 1
Clarity	Confident, audible	Audible, mostly clear	Audible only in front	Mostly inaudible	× 0.5

Maximum: 10.

Format-specific notes

- **Shopkeeper’s quick-change chart:** the chart itself is the artefact; the 3-min presentation walks through one real scenario, end-to-end, with pretend currency. The chart should be usable WITHOUT the presenter present.
- **60-second teaching video:** the video itself is the artefact; the 3-min presentation explains design choices (“we chose this example because...”). The video must show at least one full Nikhilam subtraction with verification.
- **Two’s-complement explainer poster:** the poster is the artefact; the 3-min presentation walks the audience through the binary-base 2 parallel. Should reference both Tirthaji and a CS source.

Honesty bonus (+2)

Up to 2 bonus marks for projects that explicitly address the historicity question — i.e. correctly state that the method is in Tirthaji (1965) and that the deeper Vedic provenance is unverified. This is rewarded because it models the intellectual stance we want students to take.

(Bonus capped: total project + presentation cannot exceed 40.)